

Inteligência Computacional

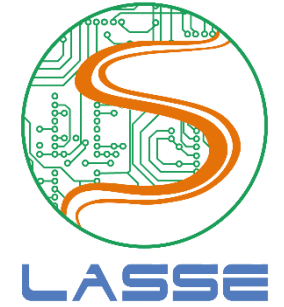
Introduction to "decision stumps" classification



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“Raw” data can often be organized as a table

- Example:



<i>Name</i>	<i>Phone</i>	<i>Weight (kg)</i>	<i>Country</i>
Trump	1 43 22885998	87	USA
Lula	55 23 591927	72	Brazil
Diaz-Canel	34 95 499402	82	Cuba
...	...	...	...



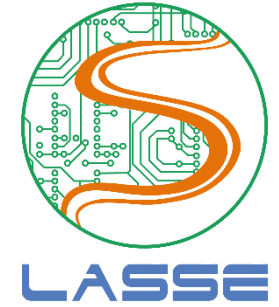
Each column is a “field”, “parameter”, “feature” or **attribute**

Each row is a “register”, “example” or **instance**

Attributes or parameters

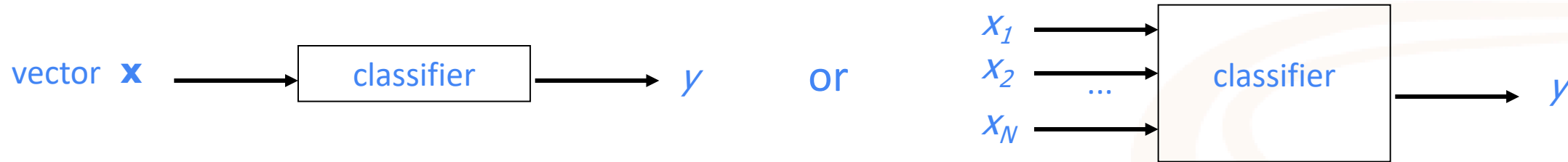
- Categories:
 - A) Nominal or Discrete, K possible values (K is an integer)
 - Binary, $K=2$. Example: Brazilian? Yes or No
 - Multiclass, $K>2$. Example: Seven rainbow colors? red, orange, yellow, green, blue, indigo and violet
 - B) Numeric
 - Real numbers. Example: weight 80.7, 23.2, 56.4, ...
 - Integer numbers. Example: age in years 1, 2, 3, ...
 - C) Strings
 - Example: Name John, Paul, Laura, etc.
 - D) Others: date, etc.
- The non-numeric attributes can be mapped to numbers
 - Example: The seven rainbow colors are mapped to integers from 1 to 7
 - Similar to the mapping of a random variable, from an “outcome” (e.g. head or tail) into a number

Classification



Problem:

- Given a vector $\mathbf{x}$ with N parameters, find its class y among Y possible labels



Fish classifier: ($Y=2$ classes)



Training set
used during →
the training stage

Length	Weight	Class y
12	3.2	
10	0.5	
14	2.8	

Test stage uses “unseen”
examples:
 $\mathbf{x} = (13, 4.2)$
 $y = ?$

Evaluation: misclassification (or error) rate

Piranha or Pirarucu?

(fishing in the Amazon rivers)



- Piranha



- Pirarucu

How classifiers are built (trained)?

- There are many automatic classifiers. Examples:
 - If / else rules and thresholds forming a “decision tree”
 - A single if/else rule with a threshold, called “decision stump”
 - K-nearest neighbors (KNN)
 - Artificial neural networks (ANN)
 - Support vector machines (SVM)
 - Naïve Bayes classifiers
 - ... and many others
- Each classifier has one or more training algorithms
- Recall: during the supervised training stage, both the parameters $\mathbf{x}$ and the correct label y is available. During test, the classifier has $\mathbf{x}$ as input and outputs its guess y_0 for the correct value of y

Simple classifiers (for two simple sets)

- Instead of using an automatic training method, let us design simple classifiers ourselves using the two simple sets below:

Training set

Length	Weight	Class y
12	3.2	0
10	0.5	1
14	2.8	0
14	2.4	0
13	1.8	1
13.8	1.5	0
11	1	1

Test set

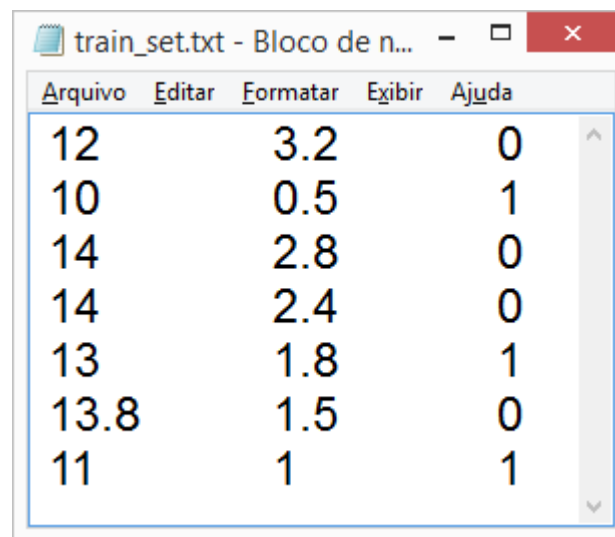
Length	Weight	Class y
13	3.1	0
9	0.8	0
12.3	1.4	1
10	2.3	1

Getting familiar with the given data

- Because there are (only) two features, it is easy to visualize the training set

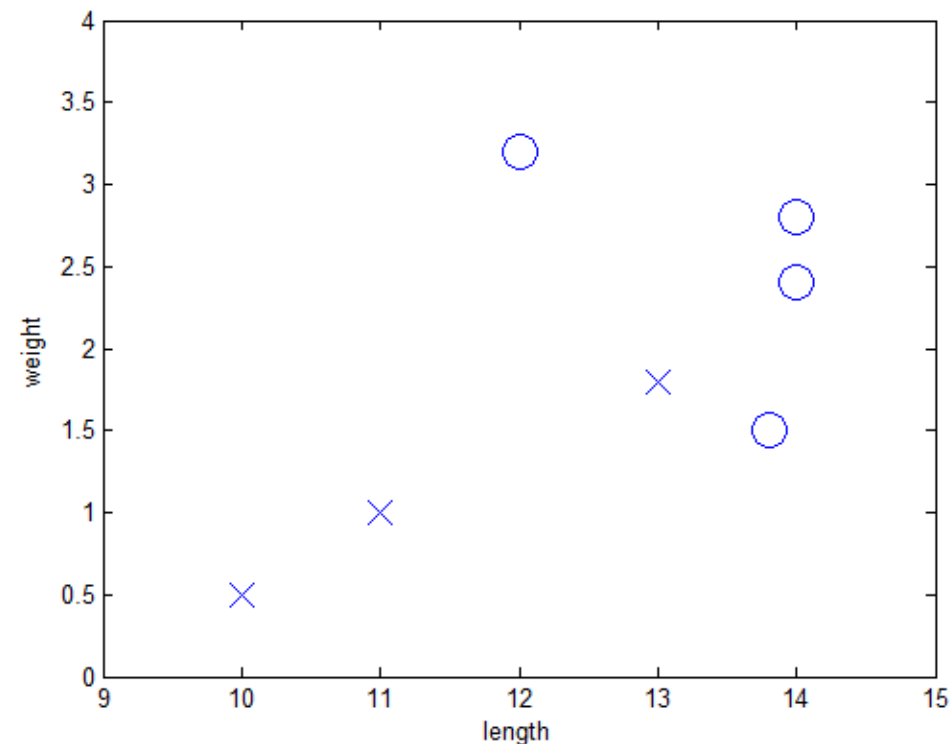
Training set

Length	Weight	Class y
12	3.2	0
10	0.5	1
14	2.8	0
14	2.4	0
13	1.8	1
13.8	1.5	0
11	1	1



```
train_set.txt - Bloco de n...  
Arquivo  Editar  Formatar  Exibir  Ajuda  
12      3.2      0  
10      0.5      1  
14      2.8      0  
14      2.4      0  
13      1.8      1  
13.8    1.5      0  
11      1        1
```

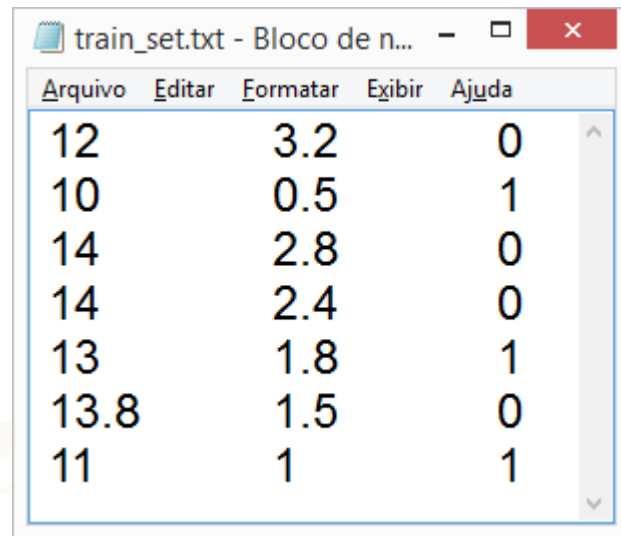
Text (ASCII) file



Manipulating the data

- Matlab / Octave code showSets.m to plot the data:

x is a 7x3 array

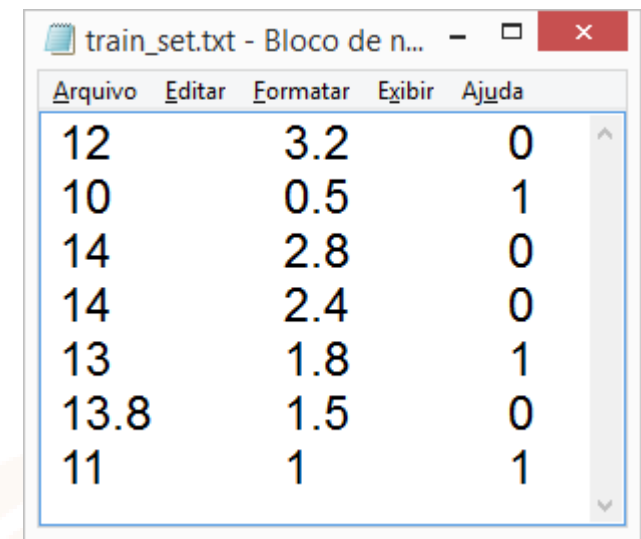


Arquivo	Editar	Formatar	Exibir	Ajuda
12		3.2		0
10		0.5		1
14		2.8		0
14		2.4		0
13		1.8		1
13.8		1.5		0
11		1		1

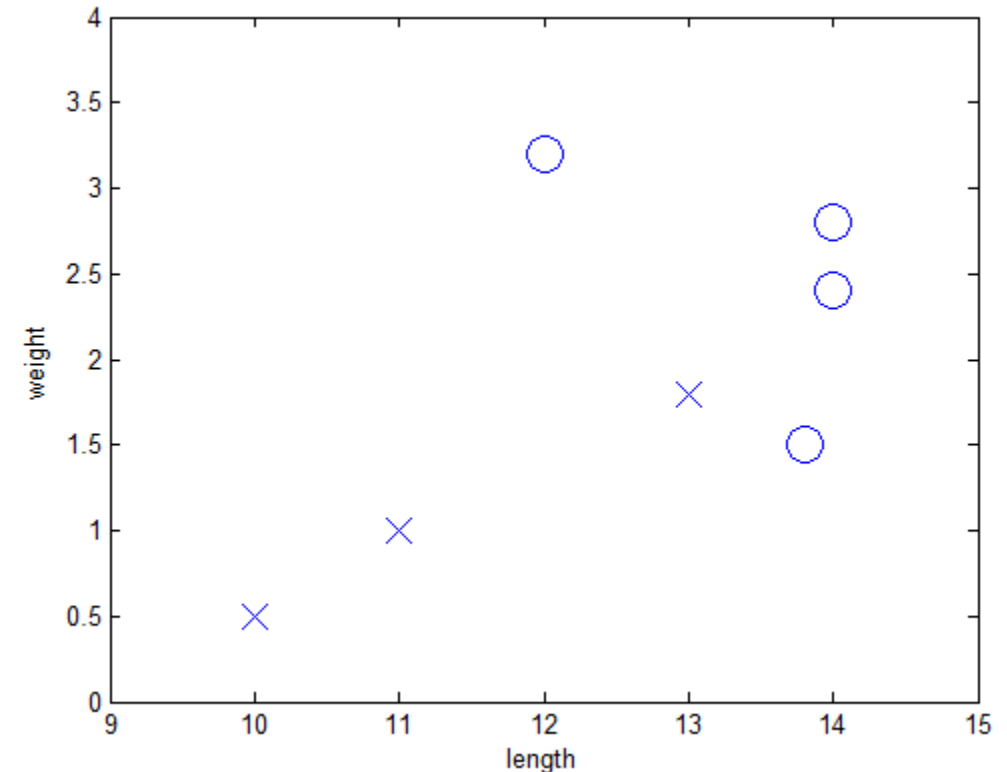
```
1 - x=load('train_set.txt'); %load from file to memory
2 - %x=load('test_set.txt'); %load from file to memory
3 - indices0=find(x(:,3)==0); %find instances with class label == 0
4 - indices1=find(x(:,3)==1); %find instances with class label == 1
5 - x0=x(indices0,1:2); %label 0: only two features, without last column
6 - x1=x(indices1,1:2); %label 1: only two features, without last column
7 - clf %clear figure
8 - plot(x0(:,1),x0(:,2),'o','MarkerSize',14) %plot label 0
9 - hold on %superimpose plots
10 - plot(x1(:,1),x1(:,2),'x','MarkerSize',14) %plot label 1
11 - axis([9 15 0 4]) %resize the x and y axes
12 - xlabel('length'), ylabel('weight')
```

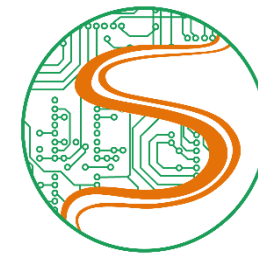
First classifier: decision stump

- Decision stump is a single if / else rule based on a chosen threshold value of a chosen feature
- First example:
 - if weight > 1
 - then class label is 0
 - This gives one error in training set!
- Another example:
 - if length < 12
 - then class label is 1
 - This also gives one error in training set!



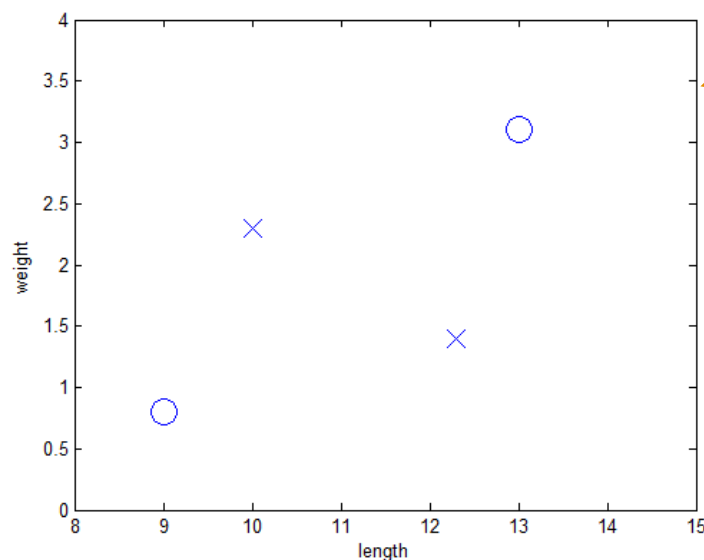
Arquivo	Editar	Formatar	Exibir	Ajuda
12	3.2	0		
10	0.5	1		
14	2.8	0		
14	2.4	0		
13	1.8	1		
13.8	1.5	0		
11	1	1		





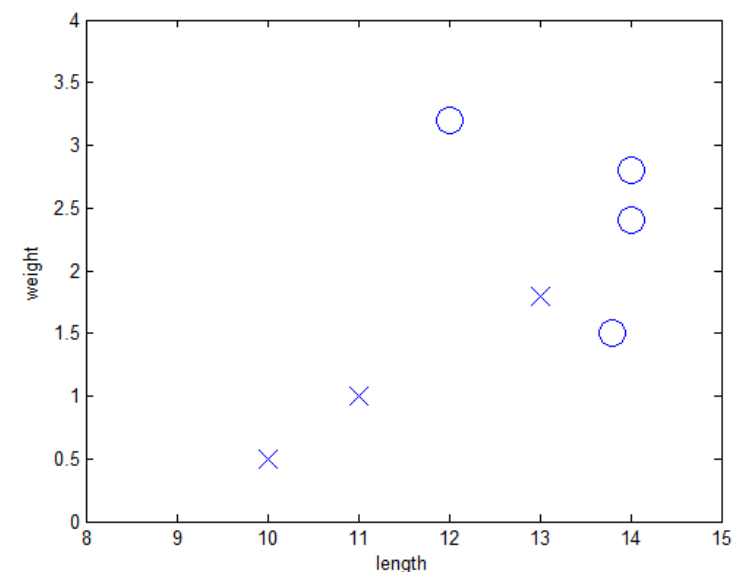
Testing our decision stump

- Example:
 - if $\text{weight} > 1$
 - then class label is 0
 - Else
 - then class label is 1
 - This gives one error in training set!
- But gives three errors in this (strange) test set!

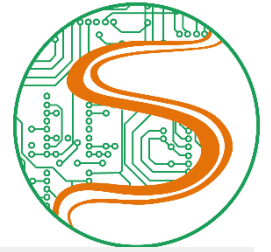


test_set.txt - Bloco de n...

Arquivo	Editar	Formatar	Exibir	Ajuda
13	3.1	0		
9	0.8	0		
12.3	1.4	1		
10	2.3	1		



Implementing (and testing): decisionStump.m



```
1 - x=load('test_set.txt'); %load from file to memory
2 - [N,M]=size(x); %N is the number of examples and M=3
3 - numberOfErrors=0; %count number of errors
4 - for i=1:N %go over all test set
5 -     exampleFeatures = x(i,1:2);%extract only features (discard label)
6 -     %decision stump: if weight > 1 then class label is 0
7 -     thisWeight = exampleFeatures(2); %chose feature
8 -     if thisWeight > 1
9 -         classifierOutput = 0
10 -    else
11 -        classifierOutput = 1
12 -    end
13 -    if classifierOutput ~= x(i,3)
14 -        numberOfErrors = numberOfErrors + 1; %update number of errors
15 -    end
16 - end
17 - display(['Error rate (%) = ' num2str(100*numberOfErrors/N)])
```